

Product Requirements Specification (PRS)

1. Product Overview

- **Product Name:** *TerraVista*
 - **One-line Description:** An intelligent, vision-powered garden management assistant that autonomously monitors plant health, predicts maintenance needs, and optimizes growth using AI.
 - **Target Users:** Home gardeners, landscaping professionals, smart home enthusiasts, and agricultural researchers.
 - **Value Proposition:** Reduces manual effort and guesswork in garden upkeep by combining real-time visual analysis and LLM-driven insights for plant care, disease prevention, and harvest optimization.
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2. Problem Statement

- **Current Gap:** Garden maintenance requires constant attention, expert knowledge, and manual monitoring for plant health, disease, watering, and harvesting schedules.
 - **Why Existing Solutions Fall Short:** Most smart garden tools are limited to moisture sensors or fixed reminders and lack holistic understanding of visual cues, environmental data, and contextual reasoning.
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3. Objectives

- Develop an AI system that autonomously observes and interprets garden conditions.
 - Detect plant species, health issues, and growth metrics through computer vision.
 - Generate actionable care recommendations via an LLM and RAG system.
 - Integrate environmental data (weather, soil, light) for predictive care.
 - Evaluate success by system accuracy, usefulness of recommendations, and reduced manual intervention.
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4. Scope

- **In Scope:**
 - Plant recognition and disease detection via CCTV feed.
 - LLM-based recommendation engine (fertilizing, watering, harvesting).
 - Weather API integration for predictive planting and maintenance.
 - Dashboard or chat-based interface for user interaction.
- **Out of Scope:**
 - Physical robotic automation (e.g., robotic mower or sprayer).
 - Complex multi-camera 3D plant mapping (for later phases).

5. Core (Functional) Features

Feature	Description	User Flow	AI/ML Role	Acceptance Criteria
Plant Recognition	Identifies plant species from CCTV feed	Camera → Detection → Plant label	Vision model (YOLOv8 or ResNet)	≥90% accuracy on known plants
Disease Detection	Flags infected or unhealthy plants	Image → Classification → Health status	CNN/ViT-based classifier	≥85% precision on common diseases
Growth Monitoring	Estimates grass length or plant maturity	Frame analysis → Measurement	Vision regression model	Accurate within ±10%
LLM Recommendation Engine	Suggests next best actions	Inputs (vision, forecast) → LLM reasoning → Advice	LLM + RAG	Contextually relevant, user-approved suggestions
Predictive Planner	Suggests planting/harvest schedule	Weather + growth + species → Forecast	Predictive modeling	≥80% accuracy in time-to-harvest predictions

6. Data Requirements

- **Sources:** CCTV footage, plant databases (PlantVillage, Kaggle datasets), weather APIs, soil/environmental sensor data (optional).
 - **Data Size & Quality:** 10k–50k labeled images for training; high-quality daytime CCTV footage for inference.
 - **Privacy:** Garden-level data only; no personally identifiable data collected.
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7. System Behavior

- **Inputs:** Real-time CCTV feed, weather forecast data, plant knowledge base.
 - **Outputs:** Plant health status, maintenance alerts, and care recommendations.
 - **Performance Expectations:**
 - Detection latency < 2s per frame.
 - Health prediction accuracy > 85%.
 - System uptime > 95%.
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8. Constraints

- **Technical:** Limited camera resolution; ambient lighting variations; compute limits on local hardware.

- **Regulatory:** None (non-personal data).
 - **Ethical:** AI transparency; avoid biased plant datasets limited to specific species.
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9. Metrics & Evaluation

- **Metrics:**
 - Plant recognition accuracy.
 - **Targets:**
 - Accuracy $\geq 90\%$, Latency $\leq 2s$, Satisfaction $\geq 4.5/5$.
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10. Timeline & Priorities

- **Phase 1 (MVP):**
 - Plant recognition + basic disease detection.
 - LLM text-based advice system.
 - **Phase 2:**
 - Predictive maintenance planner.
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